

OTR Recommended Spare Parts - Excel Analysis Report

Executive Summary

This Excel file is a **spare parts pricing calculator/catalog** for OTR (Off-The-Road) equipment manufactured by Eagle. It lists recommended spare parts for different equipment models with quantities, prices, and totals.

File: OTR Recommended Spare Parts.xlsx

Sheets: 1 (Sheet1)

Data Range: A1:E33

Purpose: Calculate total cost of recommended spare parts for multiple equipment types



Overall Structure

The spreadsheet is organized into **four equipment categories**, each listing spare parts with their quantities, unit prices, and totals:

1. **OTR Debeader** (5 parts)
2. **Punch Cutter II** (4 parts)
3. **Titan II** (4 parts)
4. **Interchangeable Parts** (1 part)

Each section includes:

- Equipment name/category header
- Column headers (Part, Quantity, Price Per Unit, Total)
- List of parts with their details
- Optional subtotal row

The spreadsheet ends with a **Grand Total** that sums all items.

Sheet Structure Details

Sheet: Sheet1

Row Range	Section	Description
Row 1	Title	"OTR Recommended Spare Parts"
Row 2	Blank	Separator
Rows 3-9	OTR Debeader	Header (row 3), 5 parts (rows 4-8), Subtotal (row 9)
Rows 10-11	Blank	Separator
Rows 12-17	Punch Cutter II	Header (row 12), 4 parts (rows 13-16), Subtotal (row 17)
Rows 18-19	Blank	Separator
Rows 20-24	Titan II	Header (row 20), 4 parts (rows 21-24), NO subtotal
Rows 25-27	Blank	Separator
Row 28	Interchangeable Parts	Header + 1 part (same row), NO subtotal
Rows 29-30	Blank	Separator
Row 31	Grand Total	Final sum
Rows 32-33	Blank	Extra space

1234

Column Layout

Column	Header	Data Type	Purpose
A	(Equipment Name)	Text	Equipment category name (merged with section header)
B	Part	Text	Part name/description
C	Quantity	Integer	Number of units needed
D	Price Per Unit	Currency (\$)	Unit price in USD
E	Total	Currency (\$)	Line item total (Quantity × Price Per Unit)

Input Cells (User Parameters)

Primary Inputs (14 parts total)

OTR Debeader (Rows 4-8)

Row	Part Name	Quantity Input	Price Input
4	Hydraulic Filter	C4	D4
5	Limit Switch	C5	D5
6	Cutter Blade	C6	D6
7	Guide Roller & Bushing	C7	D7
8	Guide Roller Pin	C8	D8

Punch Cutter II (Rows 13-16)

Row	Part Name	Quantity Input	Price Input
13	Hydraulic Filter	C13	D13
14	Sensory Rotary Encoder	C14	D14
15	Limit Switch	C15	D15
16	Punch Blade	C16	D16

Titan II (Rows 21-24)

Row	Part Name	Quantity Input	Price Input
21	Hydraulic Filter	C21	D21
22	Cutter Blade	C22	D22
23	Guide Roller & Bushing	C23	D23
24	Guide Roller Pin	C24	D24

Interchangeable Parts (Row 28)

Row	Part Name	Quantity Input	Price Input
28	PQ Control	C28	D28

Input Cell Characteristics

- **Quantity Cells (Column C):** Integer values (typically 1-2)
- **Price Cells (Column D):** Decimal currency values in USD
- **No data validation** rules currently applied
- **No constraints** or ranges defined



Output Cells (Calculated Results)

Line Item Totals (Column E)

Current State: ⚠ **HARDCODED VALUES** (Not formulas!)

All line item totals in column E are currently hardcoded numbers, not formulas. They should be calculated as:

Total (E) = Quantity (C) × Price Per Unit (D)

Affected Rows:

- E4, E5, E6, E7, E8 (OTR Debeader items)
- E13, E14, E15, E16 (Punch Cutter II items)
- E21, E22, E23, E24 (Titan II items)
- E28 (Interchangeable Parts)

Subtotals (Formulas)

Cell	Formula	Description	Current Value
E9	=SUM(E4:E8)	OTR Debeader Sub-total	\$2,821.81
E17	=SUM(E13:E16)	Punch Cutter II Sub-total	\$3,201.97

Note: Titan II and Interchangeable Parts sections do **NOT** have subtotals.

Grand Total (Formula)

Cell	Formula	Description	Current Value
E31	=SUM(E4:E30)	Grand Total of All Parts	\$19,000.02

Formula Logic: Sums all values in E4:E30, which includes:

- All line item totals
- Section subtotals (E9, E17)
- Empty cells (rows 10-11, 18-19, 25-27, 29-30)

 **Issue:** This creates **double-counting** because subtotals (E9, E17) are already sums of their respective line items, which are also included in the E4:E30 range.

Data Quality Issues Found

Issue 1: Hardcoded Line Totals

Problem: Line item totals are not calculated with formulas

Impact: If quantities or prices change, totals won't update automatically

Recommendation: Replace all line item totals with formulas: =C*D

Issue 2: Calculation Discrepancies

Several line items have incorrect totals:

Row	Part	Expected	Actual	Difference
6	Cutter Blade (OTR Debeader)	\$1,214.46	\$1,194.48	-\$19.98
16	Punch Blade (Punch Cutter II)	\$1,193.06	\$1,093.06	-\$100.00
22	Cutter Blade (Ti- tan II)	\$5,306.00	\$4,513.96	-\$792.04

Total Discrepancy: -\$912.02

These could be due to:

- Manual data entry errors
- Undocumented discounts
- Outdated prices after changes

Issue 3: Double-Counting in Grand Total

Problem: Formula `=SUM(E4:E30)` includes both line items AND subtotals

Impact: Inflates grand total incorrectly

Recommendation: Use `=E9+E17+SUM(E21:E24)+E28` to avoid double-counting

Verification:

- Current Grand Total: \$19,000.02
- Sum of all line items only: \$13,888.26
- Difference (double-counted): \$5,111.76

Issue 4: Missing Subtotals

Problem: Titan II and Interchangeable Parts sections lack subtotals

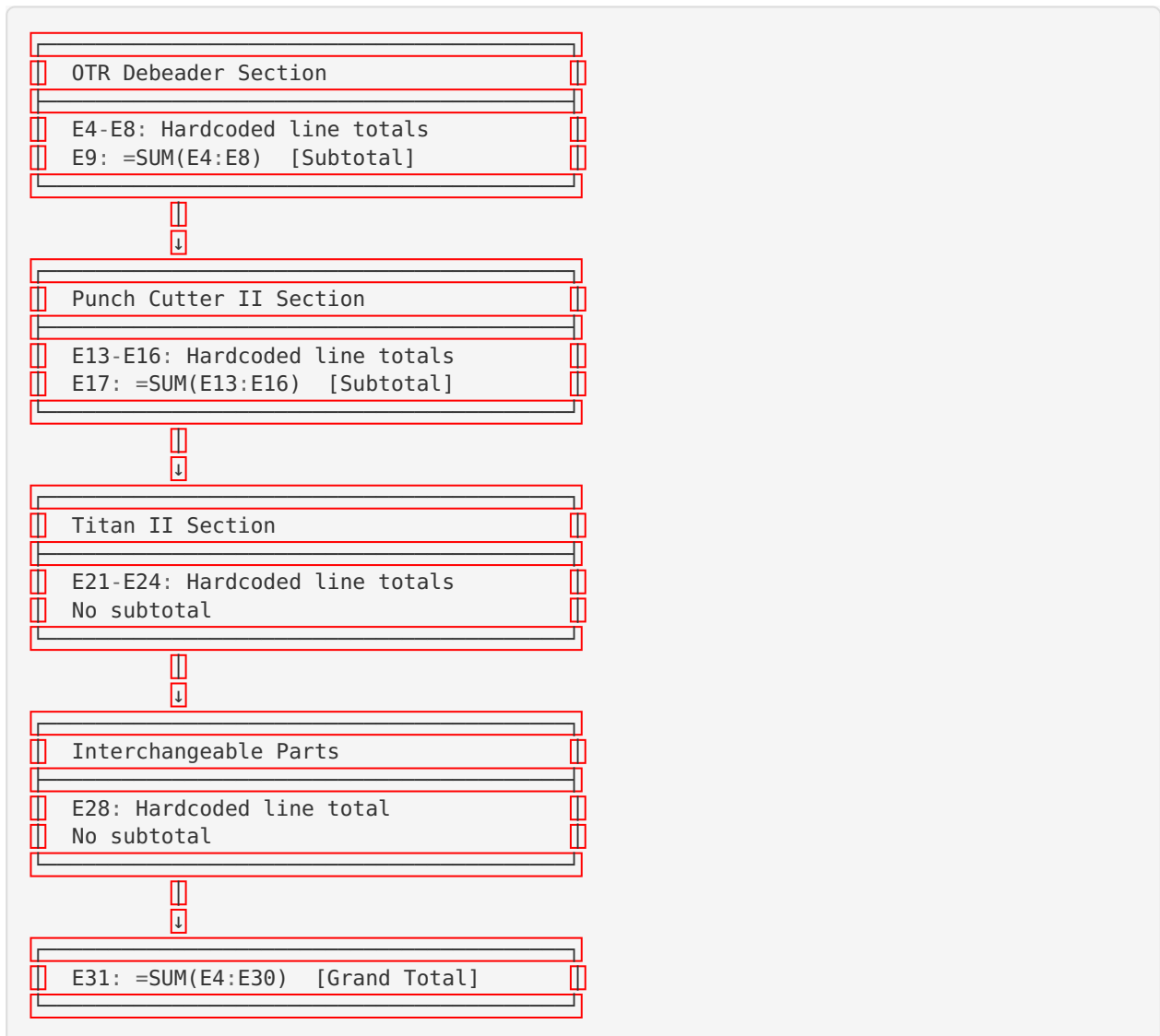
Impact: Inconsistent structure, harder to verify section totals

Recommendation: Add subtotals for consistency:

- Add Titan II subtotal in row 25: `=SUM(E21:E24)`
 - Add Interchangeable Parts label/row if needed
-

Formula Logic & Calculation Flow

Current Formula Structure



Recommended Formula Structure

Line Item Total = Quantity × Price Per Unit
 Section Subtotal = SUM(Section Line Items)
 Grand Total = SUM(All Section Subtotals)

Recommended Formulas:

Cell	Current	Recommended	Reason
E4-E8	Hardcoded	=C4*D4 to =C8*D8	Auto-calculate on input change
E9	=SUM(E4:E8)	=SUM(E4:E8)	✓ Correct
E13-E16	Hardcoded	=C13*D13 to =C16*D16	Auto-calculate
E17	=SUM(E13:E16)	=SUM(E13:E16)	✓ Correct
E21-E24	Hardcoded	=C21*D21 to =C24*D24	Auto-calculate
E25	N/A	=SUM(E21:E24)	Add Titan II subtotal
E28	Hardcoded	=C28*D28	Auto-calculate
E31	=SUM(E4:E30)	=E9+E17+E25+E28	Avoid double-counting



Formatting & Styling

Font Formatting

- **Title (A1):** Normal text
- **Section Headers (A3, A12, A20, A28):** Bold
- **Column Headers (B3-E3, etc.):** Bold
- **Data cells:** Normal text
- **Subtotals:** Normal text
- **Grand Total label (D31):** Bold

Number Formatting

- **Quantity (Column C):** General/Integer format
- **Price Per Unit (Column D):** Currency format (\$ symbol)
- **Total (Column E):** Currency format (\$ symbol)
- **Subtotals (E9, E17):** Custom format "\$#,##0.00
- **Grand Total (E31):** Custom format "\$#,##0.00

Cell Alignment

- All cells use default alignment (left for text, right for numbers)
- No special alignment rules applied

Special Features

- **No merged cells** detected
- **No data validation** rules

- **No conditional formatting**
- **No comments or notes**
- **No hidden rows or columns**

Business Logic & Rules

Pricing Rules

- **Currency:** All prices in USD (\$)
- **Quantity:** Integer values (1-2 units typical)
- **No discounts** explicitly defined
- **No tax calculations** included

Equipment Categories

1. **OTR Debeader:** Tire debanding equipment
2. **Punch Cutter II:** Tire cutting/punching equipment
3. **Titan II:** Heavy-duty tire processing equipment
4. **Interchangeable Parts:** Universal components (PQ Control)

Part Categories

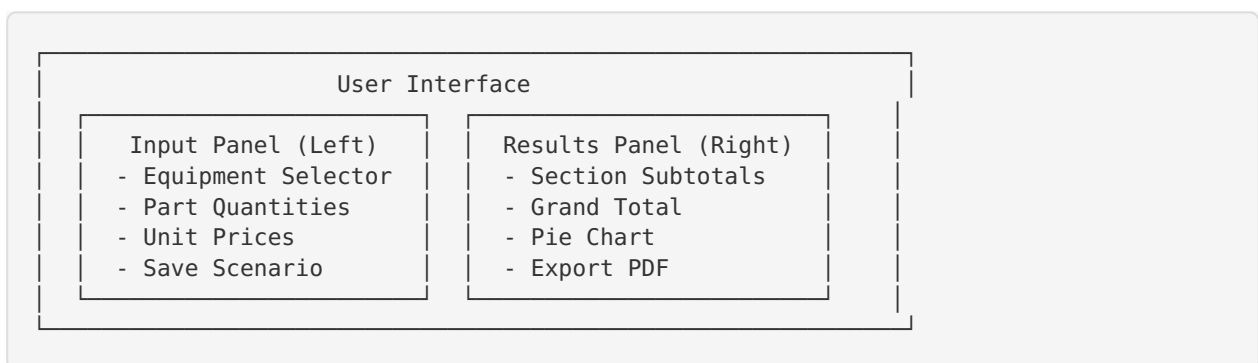
- **Hydraulic Filters:** Common across all equipment (\$181.40/unit)
- **Blades:** Cutter/Punch blades (various prices)
- **Switches:** Limit switches (\$505.23/unit)
- **Rollers & Pins:** Guide rollers, bushings, pins (various prices)
- **Electronics:** Sensory Rotary Encoder, PQ Control

Inventory Recommendations

- Most parts: 2 units recommended
- Rollers/Pins: 1 unit recommended
- Electronics: 1 unit recommended

Web Application Recommendations

Architecture Overview



Database Schema Recommendations

1. Users Table

```
users
├── id (Primary Key)
├── username
├── email
├── password_hash
├── created_at
└── updated_at
```

2. Equipment Categories Table

```
equipment_categories
├── id (Primary Key)
├── name (OTR Debeader, Punch Cutter II, etc.)
├── description
└── display_order
```

3. Parts Master Table

```
parts
├── id (Primary Key)
├── equipment_category_id (Foreign Key)
├── part_name
├── default_price
├── default_quantity
├── part_type (hydraulic, blade, electronic, etc.)
└── is_active
```

4. Calculation Scenarios Table

```
calculation_scenarios
├── id (Primary Key)
├── user_id (Foreign Key)
├── scenario_name
├── created_at
├── updated_at
└── notes
```

5. Scenario Line Items Table

```
scenario_line_items
├── id (Primary Key)
├── scenario_id (Foreign Key)
├── part_id (Foreign Key)
├── quantity
├── unit_price
├── line_total (calculated: quantity × unit_price)
└── equipment_category_id (Foreign Key)
```

6. Calculation History Table (Optional)

```

calculation_history
├── id (Primary Key)
├── user_id (Foreign Key)
├── scenario_id (Foreign Key)
├── calculation_date
├── grand_total
└── json_data (full snapshot of calculation)

```

Frontend Structure

Left Panel: Input Section

New Calculation

Scenario Name: []
[Save Scenario] [Load Scenario]

OTR DEBEADER

Part	Qty	Price
Hydraulic	[2]	[\$181.40]
Filter	[2]	[\$505.23]
...		

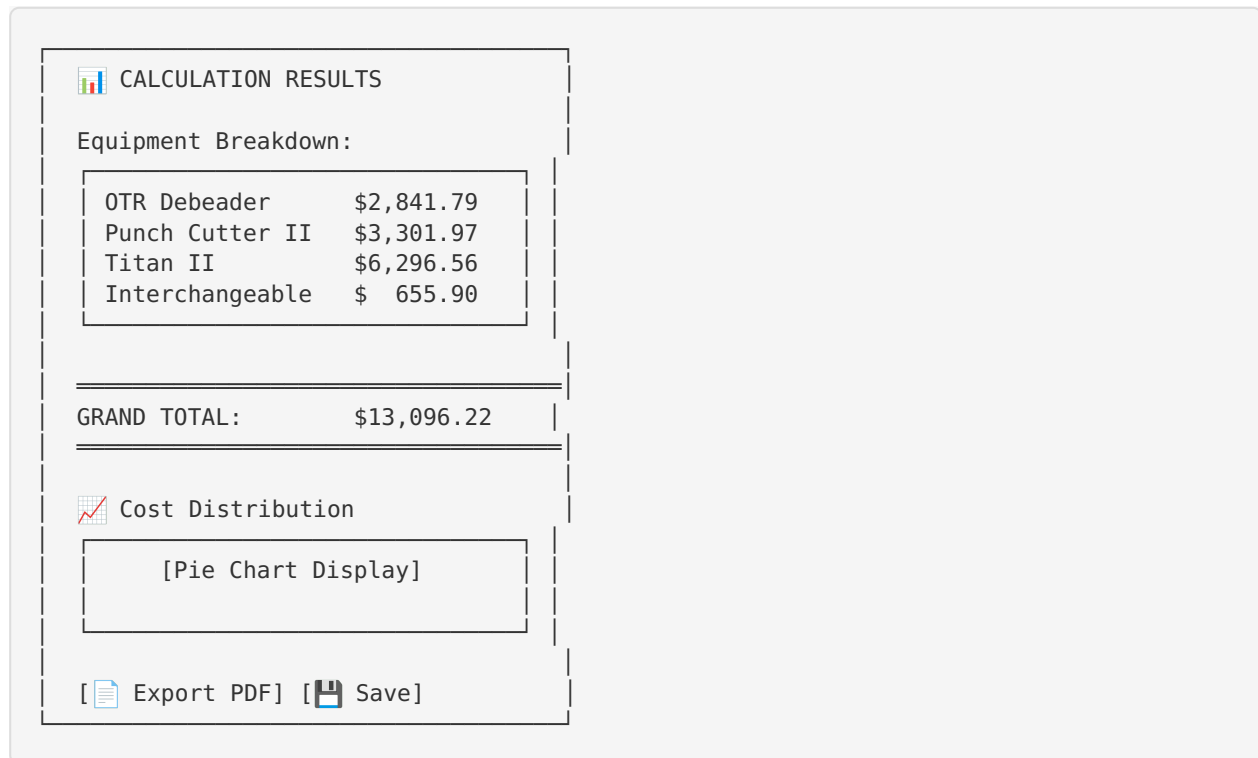
PUNCH CUTTER II

Part	Qty	Price
...		

[+ Add Custom Part]

[Calculate] [Reset]

Right Panel: Results Section



Key Features Implementation

1. Multi-User Authentication

- Use JWT tokens or session-based auth
- Implement role-based access (Admin, User)
- Password hashing (bcrypt or similar)

2. Save Calculation Scenarios

- Each user can create named scenarios
- Store all part quantities and prices
- Allow editing and updating existing scenarios
- Implement version history (optional)

3. Historical Data Storage

- Auto-save each calculation
- Provide date-based filtering
- Allow comparison between scenarios
- Export historical data to CSV/Excel

4. Pie Chart Visualization

- Show cost distribution by equipment category
- Use Chart.js or similar library
- Interactive tooltips with percentages
- Color-coded sections

5. PDF Export

- Include company logo/branding
- Show all inputs (quantities, prices)

- Display all outputs (subtotals, grand total)
- Include pie chart visualization
- Add calculation date, user name, scenario name
- Format as professional invoice/quote

Calculation Engine (JavaScript/Python)

```
// Recommended calculation logic
class SparePartsCalculator {
  calculateLineItem(quantity, unitPrice) {
    return parseFloat((quantity * unitPrice).toFixed(2));
  }

  calculateSectionTotal(lineItems) {
    return lineItems.reduce((sum, item) => {
      return sum + this.calculateLineItem(item.quantity, item.unitPrice);
    }, 0);
  }

  calculateGrandTotal(sections) {
    return sections.reduce((sum, section) => {
      return sum + section.subtotal;
    }, 0);
  }

  getCostDistribution(sections) {
    const total = this.calculateGrandTotal(sections);
    return sections.map(section => ({
      name: section.name,
      amount: section.subtotal,
      percentage: (section.subtotal / total * 100).toFixed(2)
    }));
  }
}
```

Data Validation Rules

Field	Validation	Error Message
Quantity	Integer > 0	"Quantity must be a positive integer"
Price	Decimal ≥ 0	"Price must be a positive number"
Scenario Name	String, 3-100 chars	"Scenario name must be 3-100 characters"

User Workflows

Workflow 1: New Calculation

1. User logs in
2. Navigates to "New Calculation"
3. Enters/adjusts quantities and prices
4. Clicks "Calculate"

5. Views results and pie chart
6. Optionally saves scenario
7. Exports to PDF if needed

Workflow 2: Load Existing Scenario

1. User logs in
2. Clicks "Load Scenario"
3. Selects from dropdown/list
4. System loads saved values
5. User can modify and recalculate
6. Saves as new scenario or updates existing

Workflow 3: View History

1. User logs in
2. Navigates to "History" page
3. Filters by date range
4. Views list of past calculations
5. Can reload any calculation
6. Can export historical data

Technology Stack Recommendations

Backend

- **Framework:** FastAPI (Python) or Node.js/Express
- **Database:** PostgreSQL or MySQL
- **ORM:** SQLAlchemy (Python) or Prisma (Node.js)
- **Authentication:** JWT or OAuth 2.0
- **PDF Generation:** ReportLab (Python) or PDFKit (Node.js)

Frontend

- **Framework:** React.js or Vue.js
- **UI Library:** Material-UI or Tailwind CSS
- **Charts:** Chart.js or Recharts
- **State Management:** Redux or Zustand
- **HTTP Client:** Axios

Deployment

- **Hosting:** AWS, Azure, or Heroku
- **Database:** Managed PostgreSQL service
- **File Storage:** AWS S3 (for PDF exports)
- **CDN:** CloudFront or Cloudflare

Security Considerations

- Implement HTTPS/SSL
- Sanitize all user inputs
- Use prepared statements to prevent SQL injection
- Implement rate limiting
- Add CSRF protection
- Regular security audits

- Data encryption at rest

Performance Optimizations

- Cache frequently accessed data
- Implement database indexing
- Use lazy loading for historical data
- Optimize PDF generation (background jobs)
- Implement pagination for large datasets



Summary of Findings

What This Calculator Does

- ✓ Lists recommended spare parts for 4 equipment types
- ✓ Tracks quantities and unit prices
- ✓ Calculates line totals and grand total
- ✓ Provides organized cost breakdown by equipment

What It Should Do (But Doesn't)

- ✗ Use formulas for all line item totals (currently hardcoded)
- ✗ Avoid double-counting in grand total formula
- ✗ Include consistent subtotals for all sections
- ✗ Validate/correct calculation discrepancies
- ✗ Support multi-user scenarios
- ✗ Store historical calculations
- ✗ Generate reports and visualizations

Critical Issues to Fix

1. **Replace hardcoded totals with formulas:** `=C*D` for each line item
2. **Fix grand total formula:** Use `=E9+E17+E25+E28` instead of `=SUM(E4:E30)`
3. **Add missing subtotals:** Titan II (row 25) and optionally Interchangeable Parts
4. **Verify/correct calculation errors:** Investigate discrepancies in rows 6, 16, 22

Web Application Priority Features

1. **User authentication & multi-user support** ★★★★★
2. **Save/load calculation scenarios per user** ★★★★★
3. **Automatic formula-based calculations** ★★★★★
4. **Historical data storage & retrieval** ★★★★★
5. **Pie chart visualization** ★★
6. **PDF export with full details** ★★
7. **Responsive dashboard layout** ★★
8. **Data validation & error handling** ★★



Next Steps

Phase 1: Fix Excel Issues (Optional)

- Correct all hardcoded formulas
- Fix calculation discrepancies
- Add missing subtotals
- Validate all formulas

Phase 2: Design Database Schema

- Create ER diagram
- Define all tables and relationships
- Plan data migration strategy

Phase 3: Develop Backend API

- Set up authentication system
- Create CRUD endpoints for parts, scenarios
- Implement calculation engine
- Build PDF generation service

Phase 4: Build Frontend Interface

- Create responsive dashboard layout
- Implement input forms
- Build results display with pie chart
- Integrate PDF export

Phase 5: Testing & Deployment

- Unit tests for calculations
- Integration tests for API
- User acceptance testing
- Deploy to production environment

Report Generated: January 14, 2026

Analyzed File: OTR Recommended Spare Parts.xlsx

Total Data Points: 14 parts across 4 equipment categories

Total Formulas: 3 (2 subtotals + 1 grand total)

Issues Identified: 4 critical calculation/formula issues